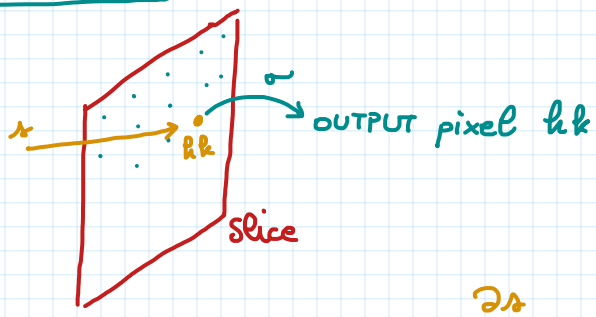
BLOCCO CONVOLUZIONALE · BACKPROPAGATION

- LOSS L_2 (MSE) : $\mathcal{L}$
- $$\frac{\partial \mathcal{L}}{\partial w_{ij}^{(\ell)}} = \delta_j^{(\ell)} \cdot x_i$$

$$\delta_j^{(\ell)} = \sum_{v=1}^{m_\ell} \delta_v^{(\ell+1)} \, w_{jv}^{(\ell+1)} \cdot \boxed{\frac{d\sigma}{d s_j^{(\ell)}} \left(s_j^{(\ell)} \right)}$$

qui c'è la CONVOLUZIONE

$$\underbrace{s_{\cancel{j}\,hk}^{(\ell)} = \left(W^{(1)} * \underline{x} + \underline{b}^{(1)} \right)_{hk}}$$



$$\frac{d\sigma}{d s_{hk}^{(\ell)}} \left(s_{hk}^{(\ell)} \right) = \sum_h \sum_k \frac{d\sigma}{ds} \left(W^{(1)} * \underline{x} + \underline{b}^{(1)} \right)_{hk} \cdot \overbrace{x_{i-h,\, j-k}^{(c)}}^{\frac{\partial s}{\partial w_{ij}}} =$$

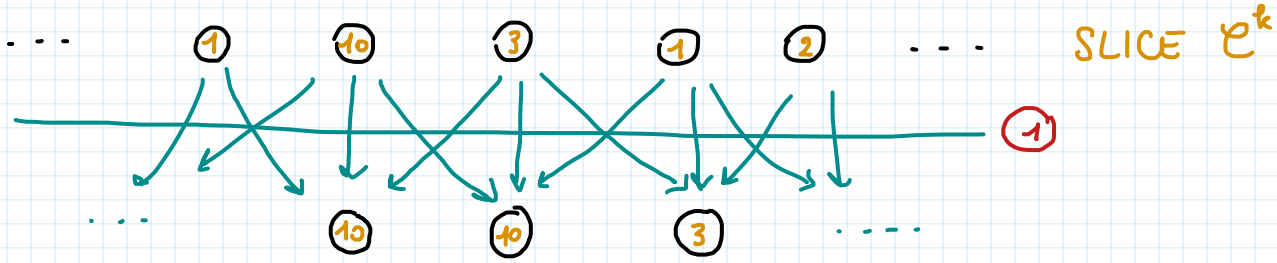
$$= \frac{d\sigma}{ds} \left(W^{(1)} * \underline{x} + b^{(1)} \right) * \underline{x}$$

$$\boxed{\frac{\partial \mathcal{L}}{\partial w_{ij}^{(\ell)}} = \sum_{v=1}^{m_\ell} \delta_v^{(\ell+1)} \, w_{jv}^{(\ell+1)} \cdot \left(\frac{d\sigma}{ds} * \underline{x} \right)_i}$$

viene implementata come CROSS-CORRELATION

LAYER di POOLING ⇝ FILTRO NON-LINEARE

Standard : MAX-POOLING



SLICE $\mathcal{C}^k$